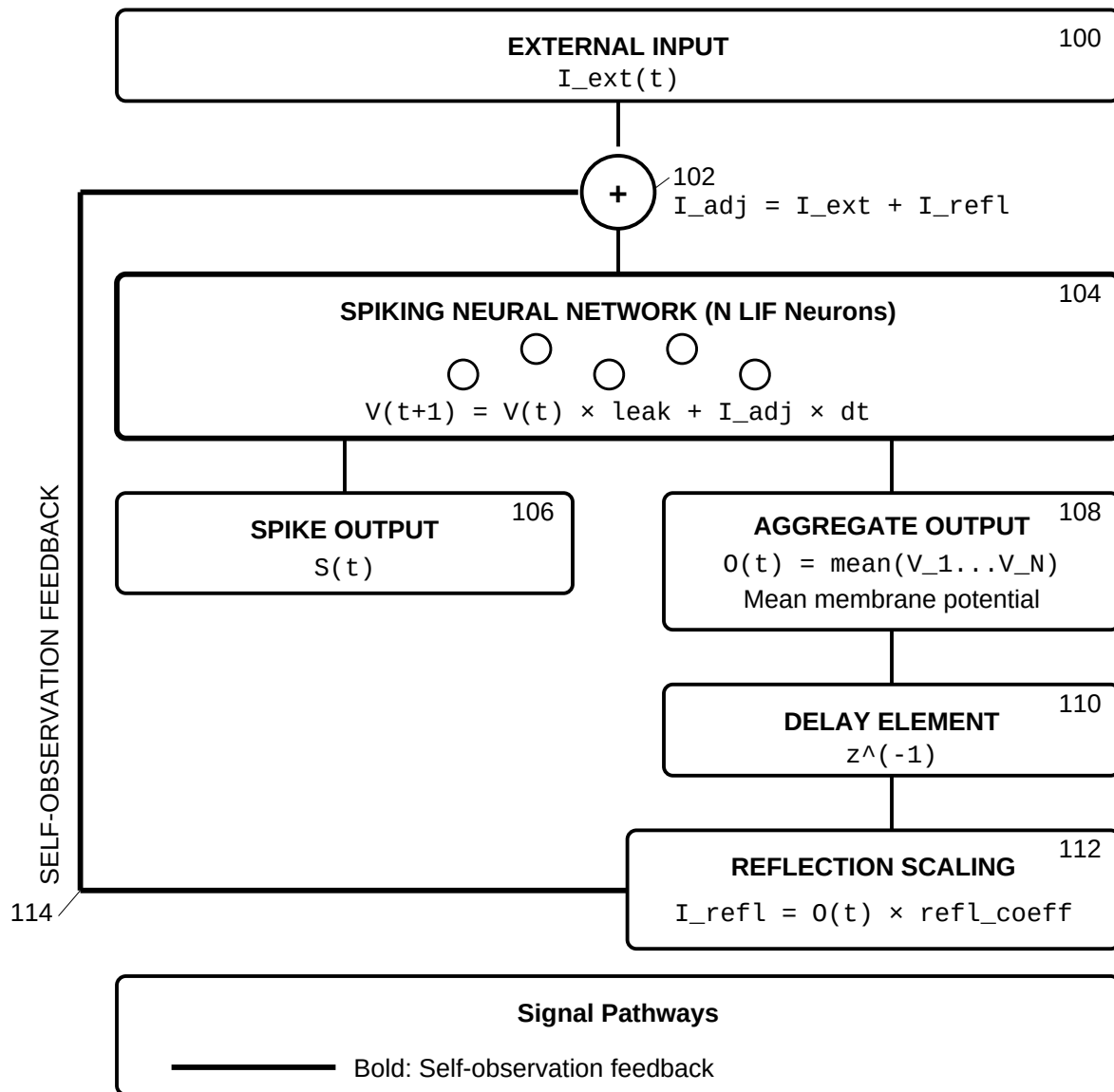


SELF-OBSERVATION FEEDBACK LOOP
Core Mechanism for Recursive Self-Observation

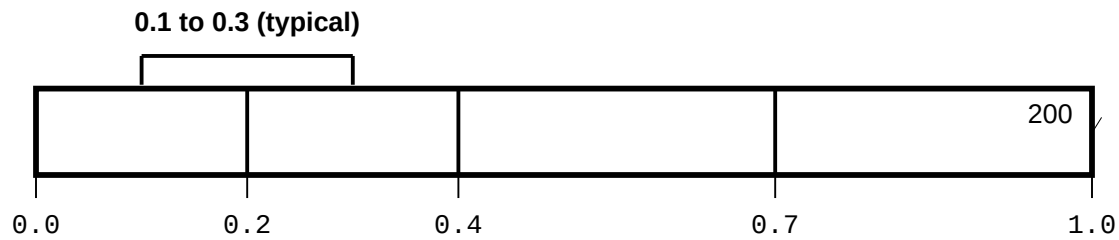


Self-observation: network observes its own prior state.
Reflection coefficient: 0.0-1.0. Operational: 0.1-0.3.

FIG. 1

REFLECTION COEFFICIENT SPECTRUM

Behavioral Effects Across Full Range (0.0 to 1.0)



Region 1 (0.0): No Self-Observation

202 ✓

Pure feedforward processing

Behavior: System responds only to external input

Region 2 (0.0 - 0.2): Minimal Self-Awareness

204 ✓

Slight influence of prior state

Behavior: Primarily externally driven

Region 3 (0.2 - 0.4): Balanced Self-Observation

OPERATIONAL RANGE 206 ✓

External input + meaningful self-reference

Default: $\text{refl_coeff} = 0.2 + \text{modulation} \times 0.1$

Region 4 (0.4 - 0.7): Self-Dominated Processing

208 ✓

Internal state dominates

Behavior: Risk of echo chamber dynamics

Region 5 (0.7 - 1.0): Self-Absorption

210 ✓

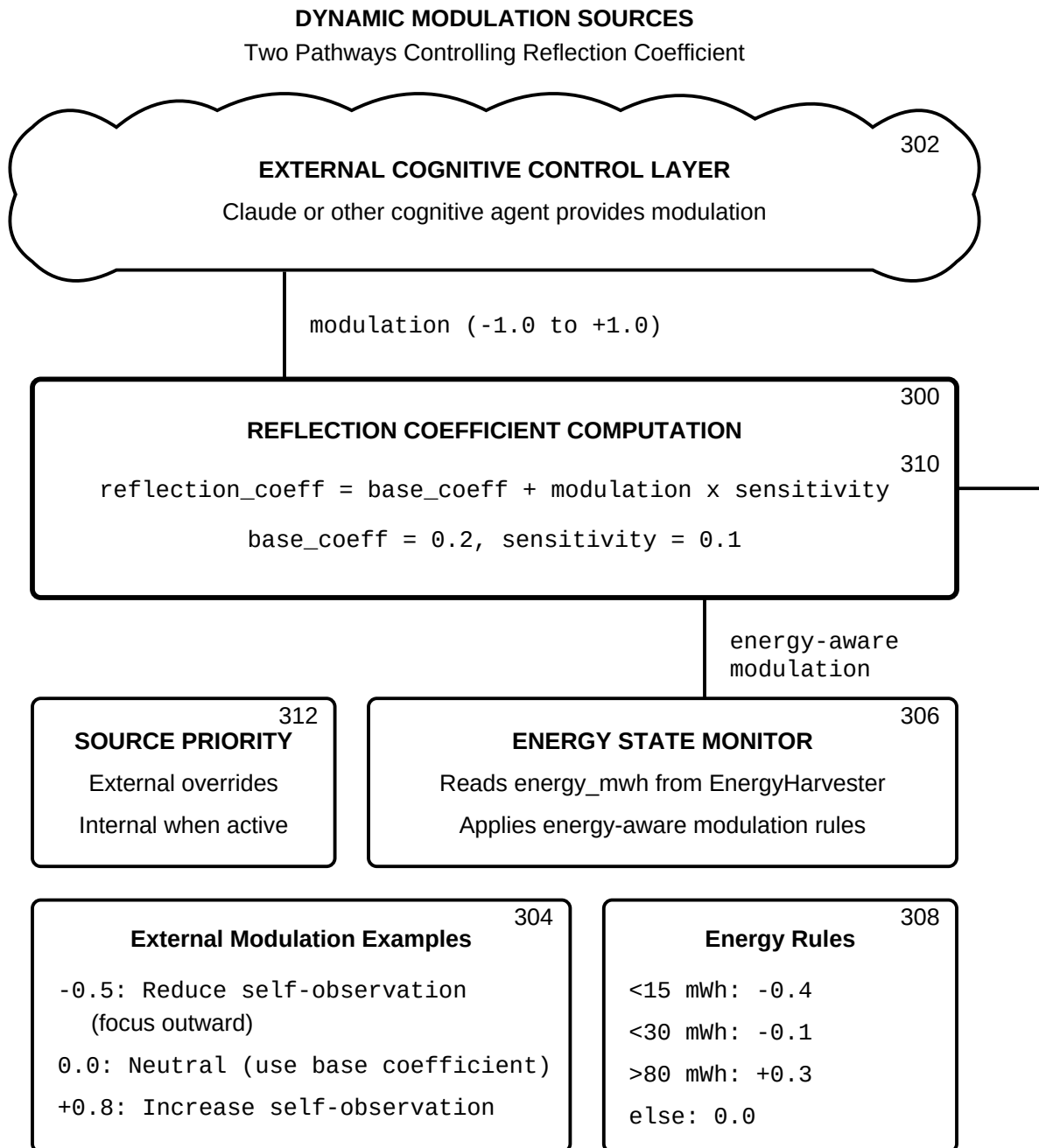
External input overwhelmed

Behavior: System locks onto its own prior state

Increasing self-referential processing

The reflection coefficient is the system's self-awareness dial

FIG. 2

**FIG. 3**

SELF-REFERENTIAL LEARNING LOOP
(STDP + Self-Observation)

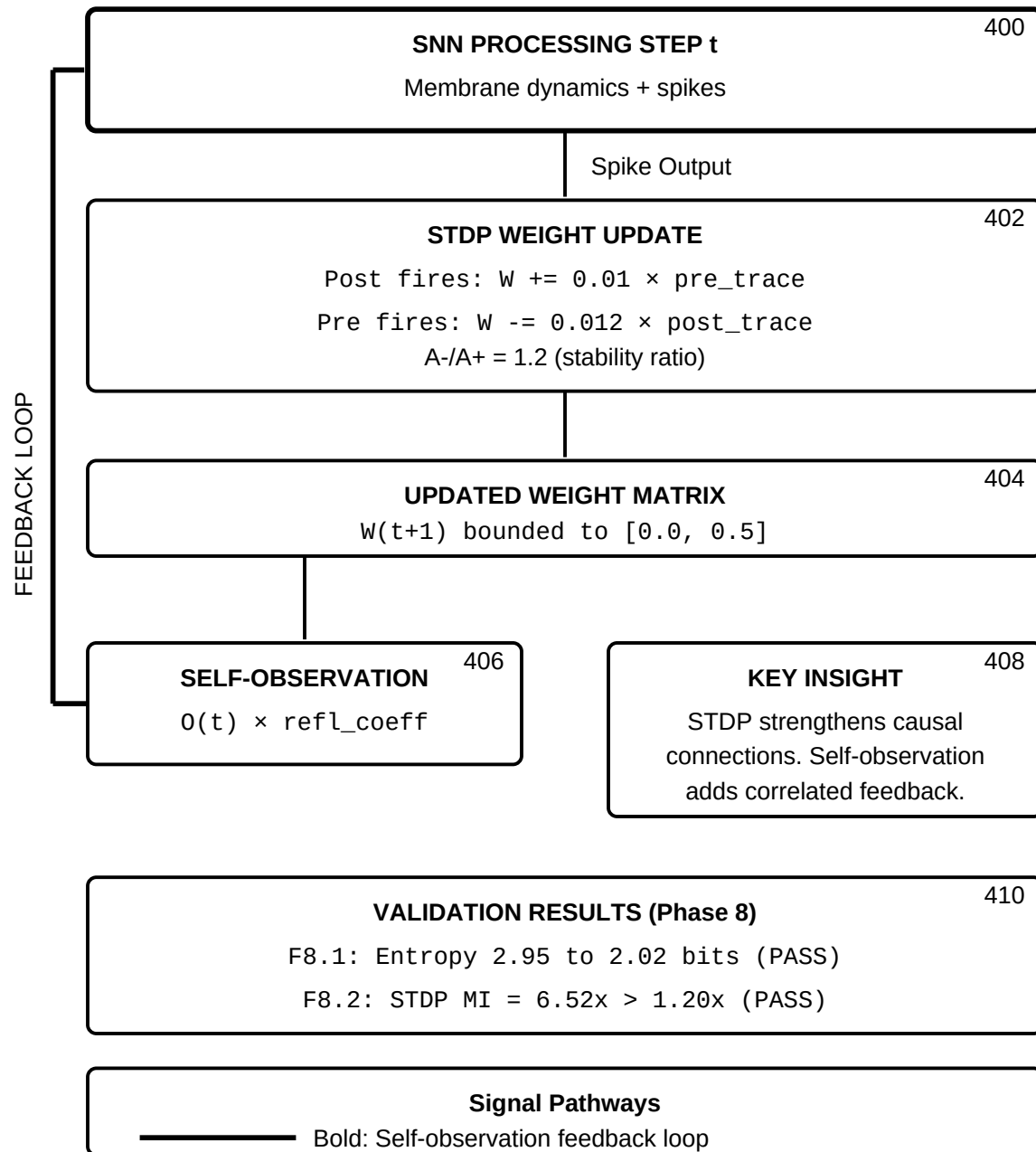
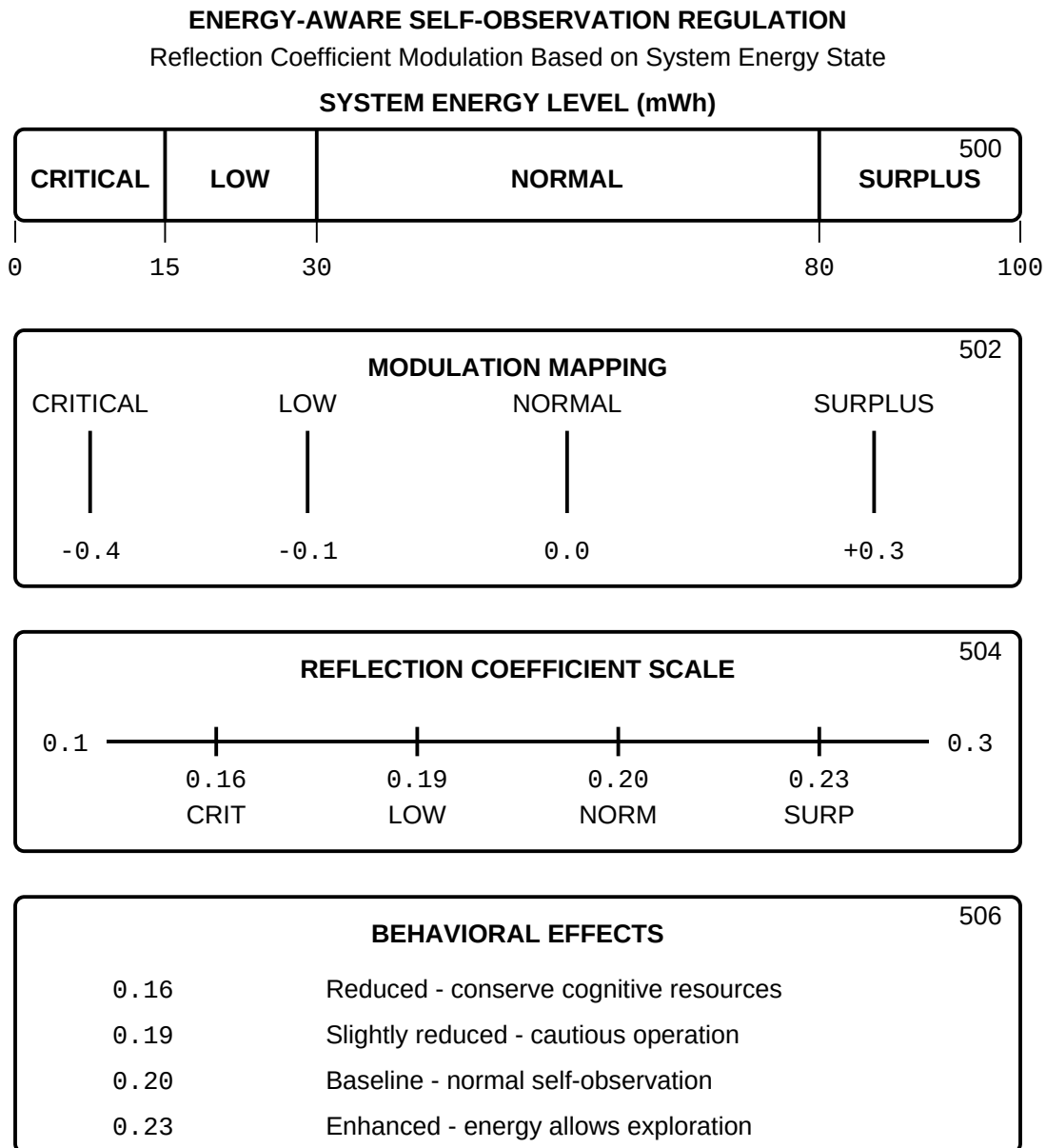


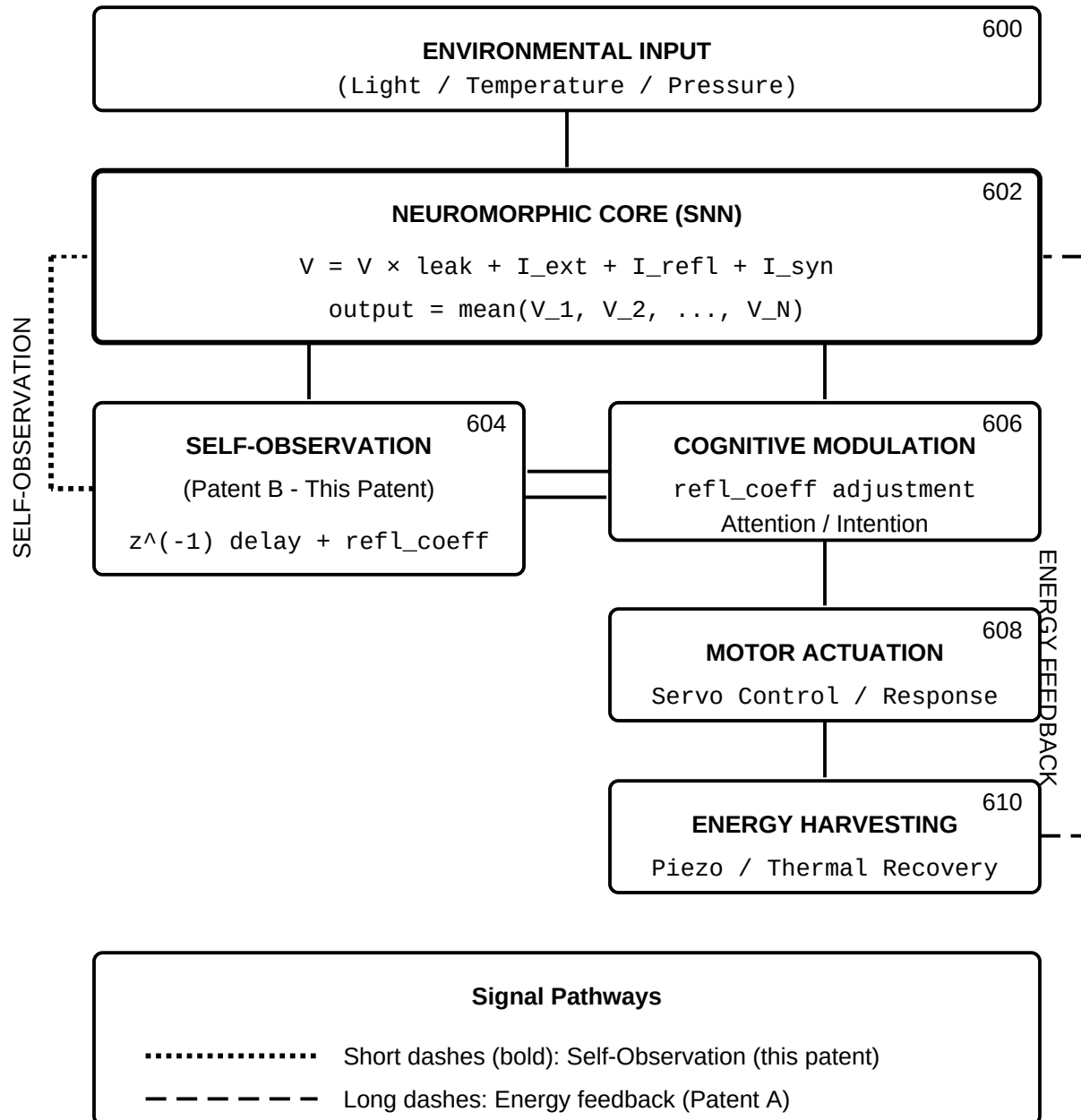
FIG. 4



Energy-aware regulation ensures efficient cognitive resource management.
Low energy reduces self-observation; surplus energy enables exploration.

FIG. 5

END-TO-END SIGNAL FLOW
Self-Observation Integration in Complete System



The self-observation pathway feeds the network's own aggregate state back as explicit input at configurable gain (0.0 to 1.0).

FIG. 6